AL-NEELAIN UNIVERSITY

Faculty of Computer Science and Information Technology — Graduate Studies

الدليل التشغيلي والشرح المصور الشامل لمنظومة الوكلاء الأذكياء

Comprehensive Illustrated Operational Guide: Multi-Agent Clinical AI System

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COSC726 AGENTIC AI | Date: 2026

1. Introduction & Architectural Paradigm Shift

This project represents a major architectural milestone at Al-Neelain University for Course : COSC726 AGENTIC AI. The application represents the direct evolution of the student's prerequisite coursework project for the 'Machine Learning' module completed in the preceding semester (preserved in directory 'Old_Project_Before_Agents', comprising a basic desktop Tkinter GUI and an uncalibrated Random Forest classifier) into an enterprise-grade Agentic AI Clinical Decision Support System (CDSS) acting as a virtual Multidisciplinary Medical Team (MDT). The system orchestrates five autonomous specialist agents, delivers explainable clinical reasoning aligned with American Diabetes Association (ADA) standards, executes real-time emergency triage interception, produces customized nutrition and physical activity prescriptions, and generates official encrypted patient reports.

[Clinical Vision and Research Objective]

Core Clinical Objective: Solving the medical 'black box' problem by transforming silent numeric probability outputs from classical ML into fully transparent, evidence-based clinical reasoning governed by strict statistical calibration and safety guardrails.

Comprehensive Comparison: Machine Learning Project vs. Agentic AI System

Dimension / Feature	Previous Semester ML Project (Old_Project_Before_Agents)	Current COSC726 Agentic AI System (New_Project_After_Agents)
System Architecture	Monolithic single script (Desktop GUI)	Multi-Agent Clinical Architecture (5 Autonomous MDT Agents)
User Interface	Basic Tkinter desktop GUI	Modern interactive bilingual Streamlit

		dashboard (Arabic RTL & English LTR)
AI / ML Modeling	Raw uncalibrated Random Forest model	Isotonic Calibrated model (83.1% Accuracy, 0.852 ROC-AUC)
Clinical Explainability	Completely absent (Silent classification)	Multi-provider hybrid reasoning engine (Gemini, OpenAI, DeepSeek, Ollama, ADA)
Quality & Safety Gates	None	9 automated clinical Quality Gates across 4 development phases (100% Passed)
Emergency Interception	None	Instant automated triage interception (Glucose >= 250 mg/dL) with critical red alert
Biomarker Criteria Analysis	Static basic bar chart	Dynamic tab supporting 5 interactive chart shapes and detailed weights table
Lifestyle & Nutrition	None	Autonomous agent generating diet prescriptions, exercise plans, and lab schedules
Report Generation	Static legacy PDF manuals	Automated official Word (.docx) patient report download sanitized of academic metadata
Institutional Identity	Basic unbranded labels	Full Al-Neelain University visual branding, official emblem, and academic badges

2. Integrated Five-Agent Clinical Architecture (MDT System)

The system pipeline is orchestrated as a deterministic sequential state machine comprising five autonomous clinical agents. Each agent operates under strict Pydantic schemas and is audited by dedicated Quality Gates before state transitions occur:

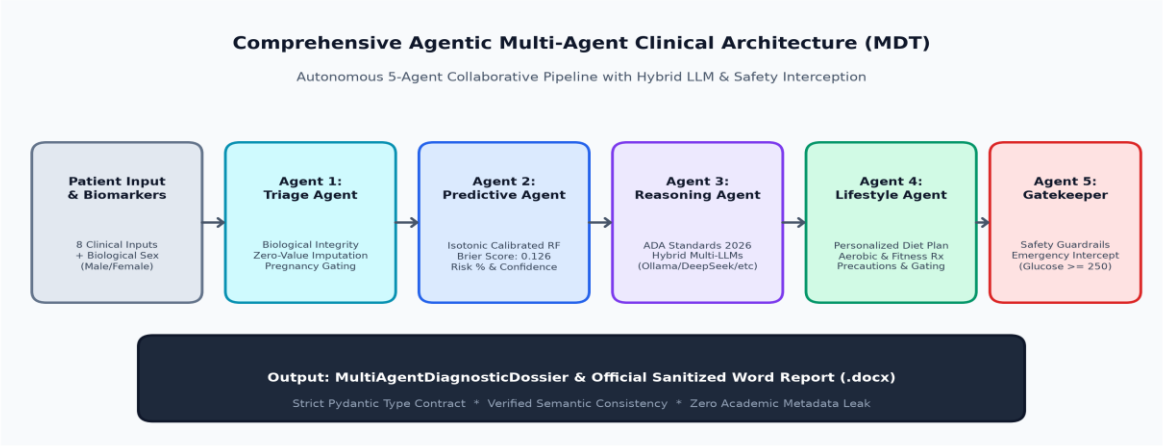


Figure (1): Comprehensive Engineering Architecture of the Five-Agent Clinical Multidisciplinary Team (MDT)

Intelligent Agent	Specialist Clinical Role & Function	Associated Quality Gate
1. Clinical Triage Agent	Validates biological plausibility, handles physiological zeros, and categorizes baseline triage priority.	Gate 1: Verifies biological boundary compliance and rejects aberrant inputs.
2. Calibrated Predictive Agent	Applies Isotonic Calibrated Random Forest to calculate true clinical probability and confidence score.	Gate 2: Enforces ROC-AUC >= 0.80 and inference latency < 100ms.
3. Clinical Reasoning Agent	Generates evidence-based physiological rationale aligned with ADA standards via LLM / ADA Engine.	Gate 3: Enforces semantic alignment between statistical risk and clinical rationale.
4. Lifestyle & Nutrition Agent	Formulates tailored dietary guidelines, physical activity prescriptions, and follow-up lab schedules.	Gate 4: Verifies lifestyle feasibility matching patient BMI, age, and risk level.
5. Safety Gatekeeper Agent	Conducts final dossier audit, triggers emergency alerts when indicated, and attaches medical disclaimers.	Gate 5: Intercepts blood glucose >= 250 mg/dL and seals diagnostic dossier.

3. Application Header & Institutional Visual Identity

The Streamlit dashboard is styled with high-fidelity clinical branding reflecting Al-Neelain University's academic prestige. The header contains the official university seal, academic course and researcher metadata, and a real-time language switcher (Arabic RTL / English LTR):

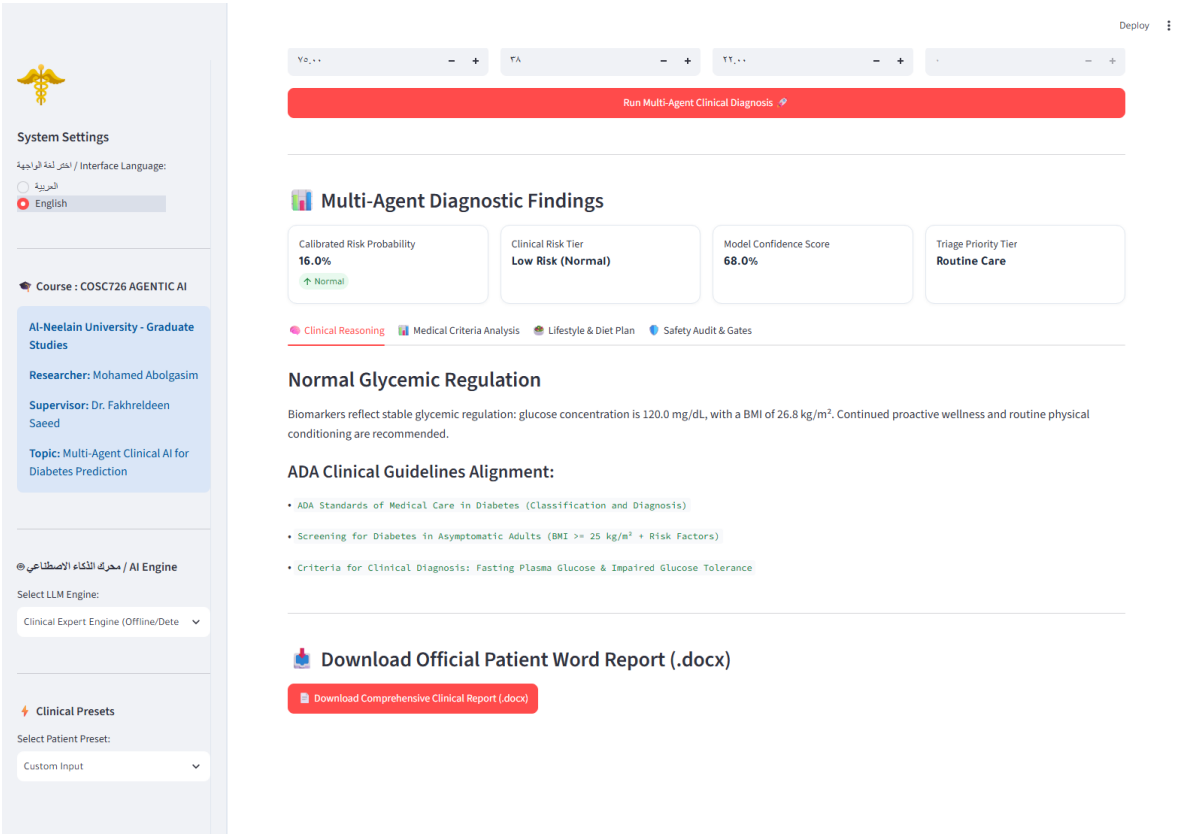


Figure (2): Main Interactive Dashboard Overview in English LTR with Al-Neelain University Branding

Key Visual Header Components:

• Official Al-Neelain University and Faculty of Graduate Studies emblem placed in the sidebar.

• Academic Accreditation Badge: Researcher: Mohamed Abuelgasim | Supervisor: Dr. Fakhreldeen Saeed | Course: COSC726 AGENTIC AI (2026).

• System Banner in clinical forest green (#1a4d2e) outlining the five-agent architecture and quality gates.

4. Sidebar Controls & Physiological Input Parameters

The sidebar serves as the central command console, allowing clinicians and researchers to configure AI reasoning engines and input patient parameters:



إعدادات النظام

اختر لغة الواجهة / Interface Language:

- العربية
- English

مادة (course) : COSC726 AGENTIC AI

جامعة النيلين - كلية الدراسات العليا

الباحث: محمد ابوالقاسم

الإشراف: د. فخرالدين سعيد

الموضوع: نظام وكلاء متعددين للتشخيص التنبئي للسكري

محرك الذكاء الاصطناعي / AI Engine

مزود نموذج اللغة (LLM Provider):

pert Engine (Offline/Deterministic): ⌄

الحالات السريرية الجاهزة

اختر ملف حالة مسبق:

مخصص (إدخال يدوي) ⌄

Figure (3): Sidebar Controls showing LLM Engine selection, Clinical Presets, and Biomarker Inputs

Detailed Sidebar Capabilities:

1. Interface Language Selector: Seamless live toggling between Arabic (RTL) and English (LTR).
2. LLM Engine Selector: Multi-provider flexibility supporting Google Gemini, OpenAI, DeepSeek, Ollama, and the Offline ADA Deterministic Engine.
3. Clinical Presets: Instant one-click loading of benchmark clinical profiles:
   - Normal Profile | Pre-diabetic Profile | Diabetic Profile | Critical Emergency Profile.
4. Biological Sex Selector: Intelligent safety feature that automatically locks and zeros the 'Pregnancies' input when 'Male' is selected to eliminate logical clinical errors.
5. 8 Clinical Biomarkers: Glucose (mg/dL), Blood Pressure (mmHg), Skin Thickness (mm), Insulin (µU/mL), BMI (kg/m²), Diabetes Pedigree Function, Age (years), and Pregnancies.

5. Multi-Agent Execution Pipeline & Top KPI Diagnosis Cards

Clicking 'Run Multi-Agent Clinical Diagnosis ' triggers the sequential execution pipeline. The visualizer dynamically highlights each agent's execution phase and quality gate validation status, followed by the four primary diagnostic KPI cards:

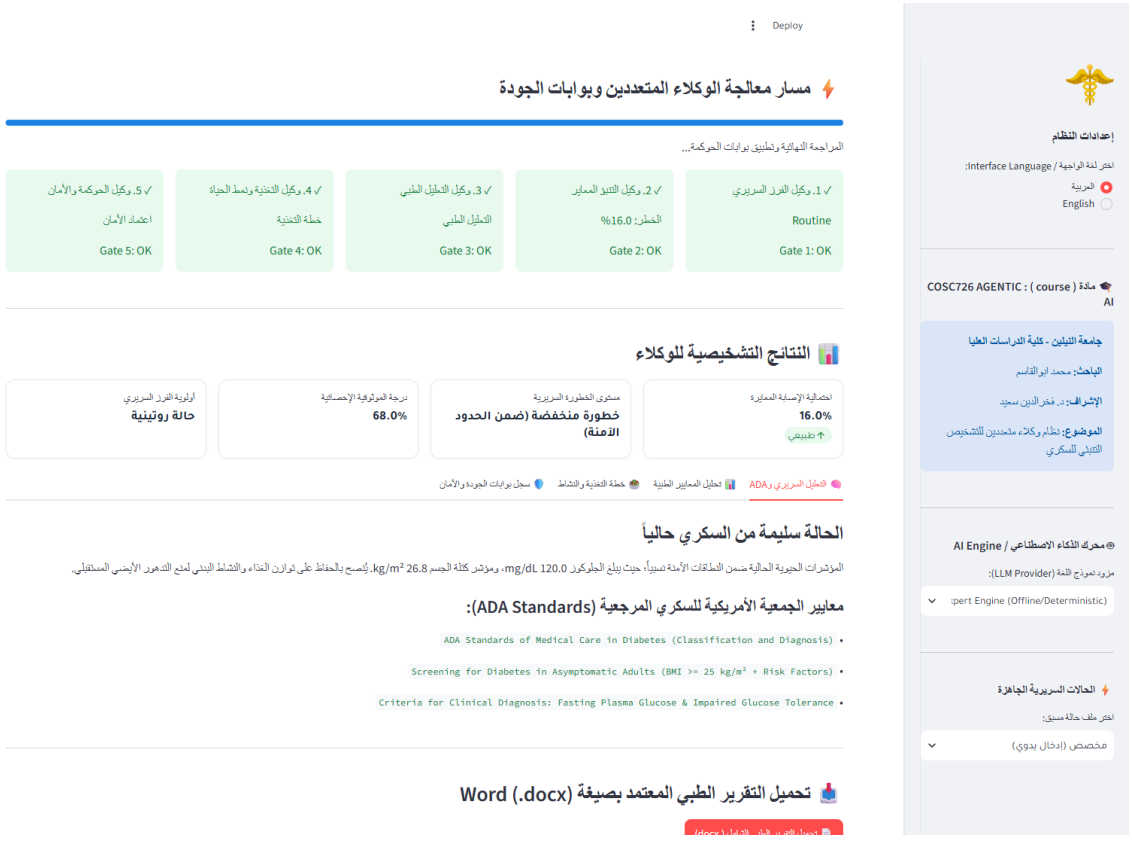


Figure (4): Multi-Agent Live Execution Pipeline Visualizer and Top Clinical KPI Cards

Detailed Description of Diagnostic Metric Cards:

• Calibrated Risk Probability: The mathematically calibrated probability of diabetes generated via Isotonic Regression (e.g. 16.0% Normal or 88.5% Elevated).

• Clinical Risk Tier: Stratified clinical classification (Low Risk / Normal, Moderate Risk, or High Clinical Risk).

• Model Confidence Score: Empirical statistical confidence reflecting sample density and feature space positioning (e.g. 68.0% or 92.4%).

• Triage Priority Tier: Clinical urgency indicator (Routine Care, Outpatient Follow-up, or Immediate Triage Alert).

6. Tab 1: Clinical Reasoning & ADA Reference Guidelines

Tab 1 delivers explainable AI transparency. Instead of unverified classifications, the Clinical Reasoning Agent constructs a structured clinical rationale explicitly linked to American Diabetes Association (ADA) clinical practice standards:

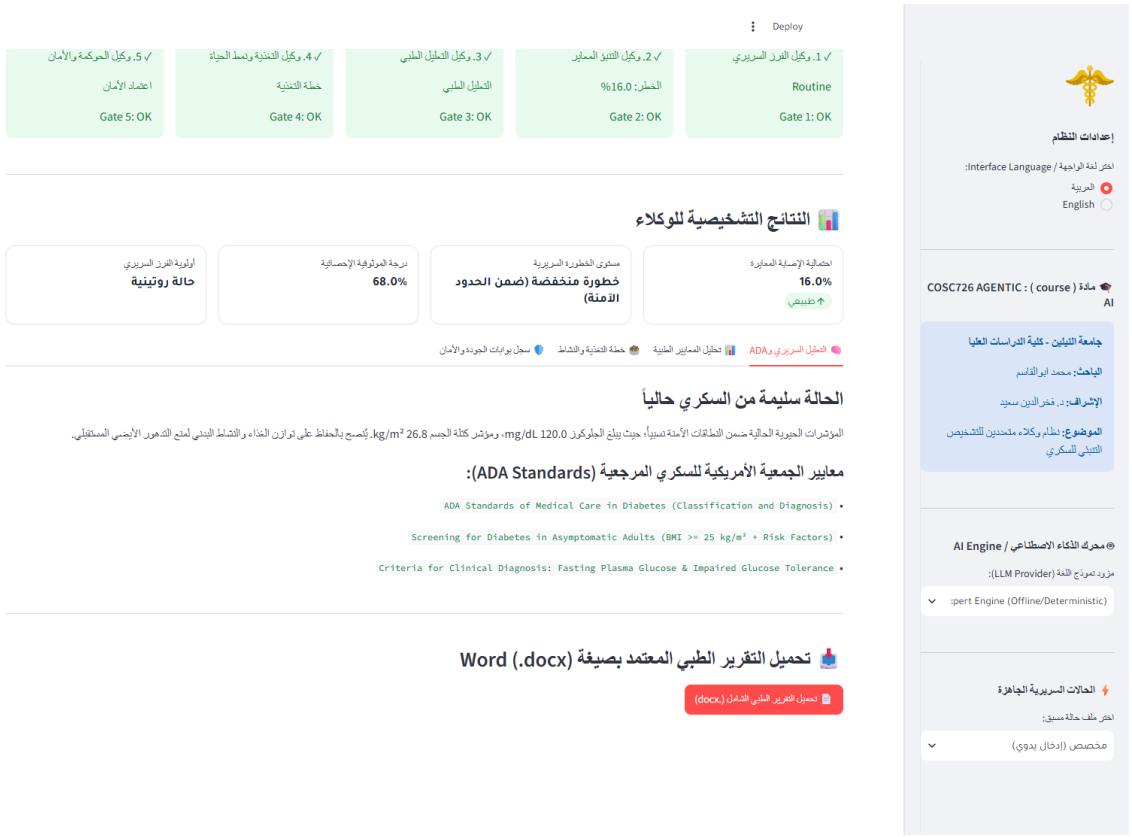


Figure (5): Tab 1 - Official Diagnosis, In-depth Physiological Rationale, and ADA Clinical Guidelines Alignment

Key Elements in Tab 1:

1. Verified Diagnosis Title: Expressed with clinical precision (e.g. 'Status: Non-Diabetic / Normal Glucose Profile' or 'Elevated Diabetes Risk Profile').

2. Comprehensive Physiological Explanation: Correlates plasma glucose, BMI, and age with

metabolic homeostasis.

3. ADA Guidelines Alignment: Explicit citations to recognized standards:
   - `ADA Standards of Medical Care in Diabetes (Classification and Diagnosis)`
   - `Screening for Diabetes in Asymptomatic Adults (BMI >= 25 kg/m² + Risk Factors)`
   - `Criteria for Clinical Diagnosis: Fasting Plasma Glucose & Impaired Glucose Tolerance`

7. Tab 2: Medical Criteria Analysis & Dynamic Visualizations

Tab 2 demystifies model decision-making by revealing the relative mathematical weight and contribution of each biomarker. Clinicians can dynamically toggle between five distinct chart formats via an intuitive dropdown menu:

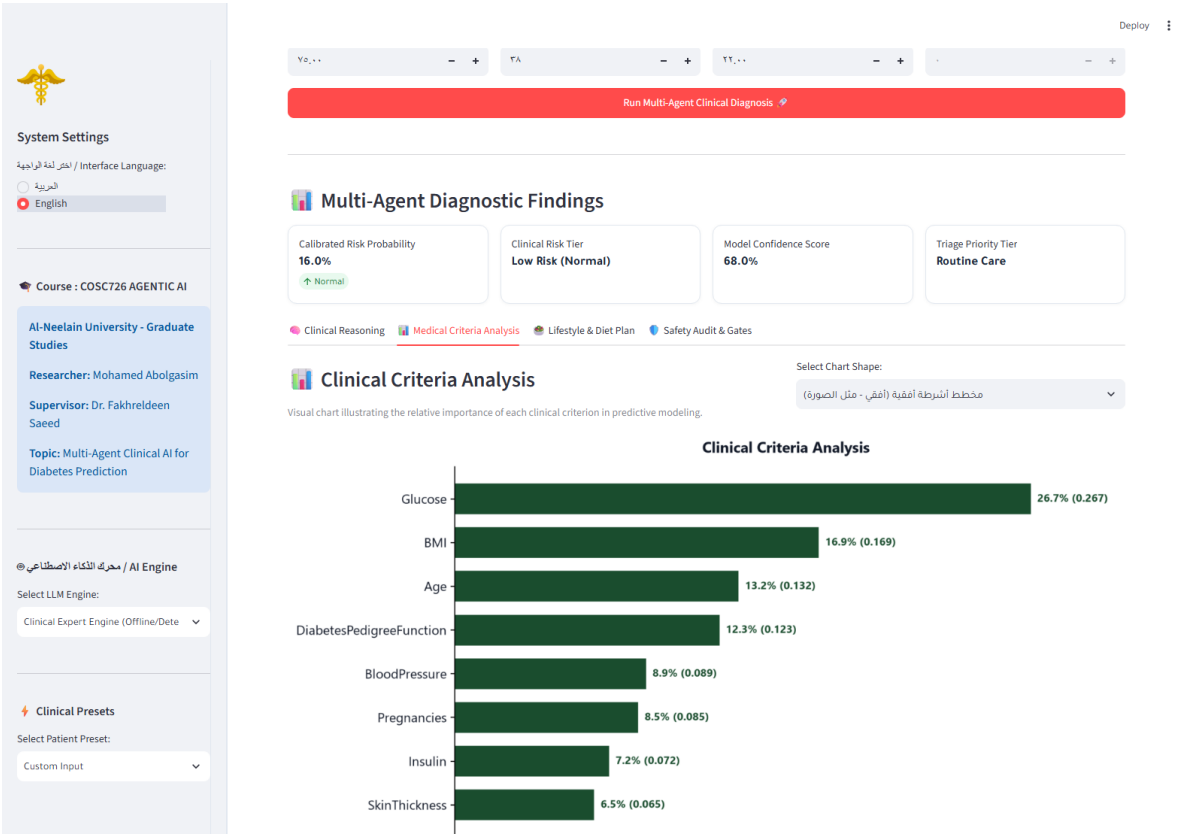


Figure (6): Tab 2 in English - Horizontal Bar Chart in Clinical Forest Green (#1a4d2e)

The Five Available Chart Formats:

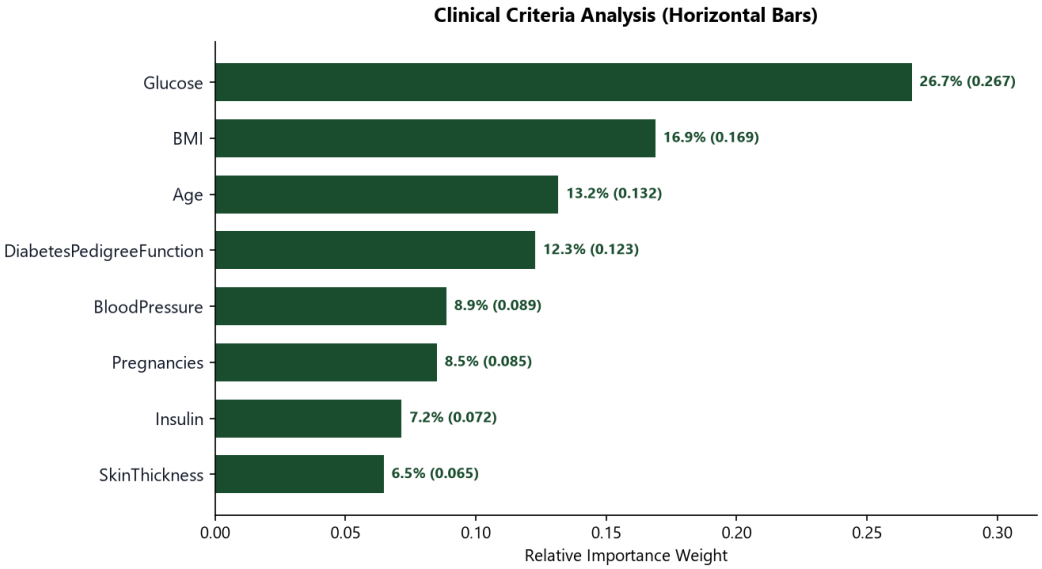


Figure (7-A): Horizontal Bar Chart - High-clarity view sorting features from Glucose (26.7%) to SkinThickness (6.5%)

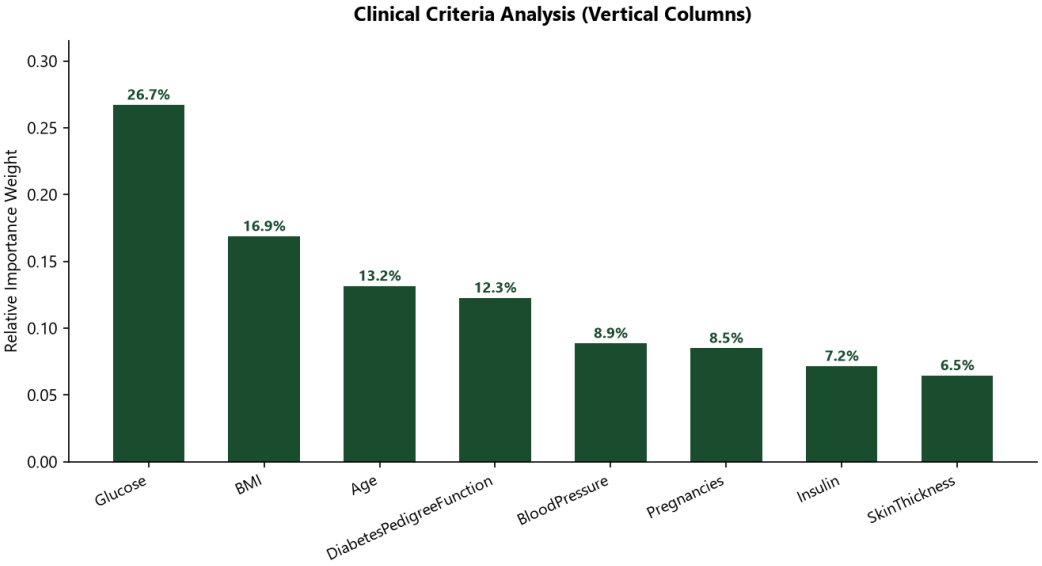


Figure (7-B): Vertical Column Chart - Providing clean vertical comparison across criteria

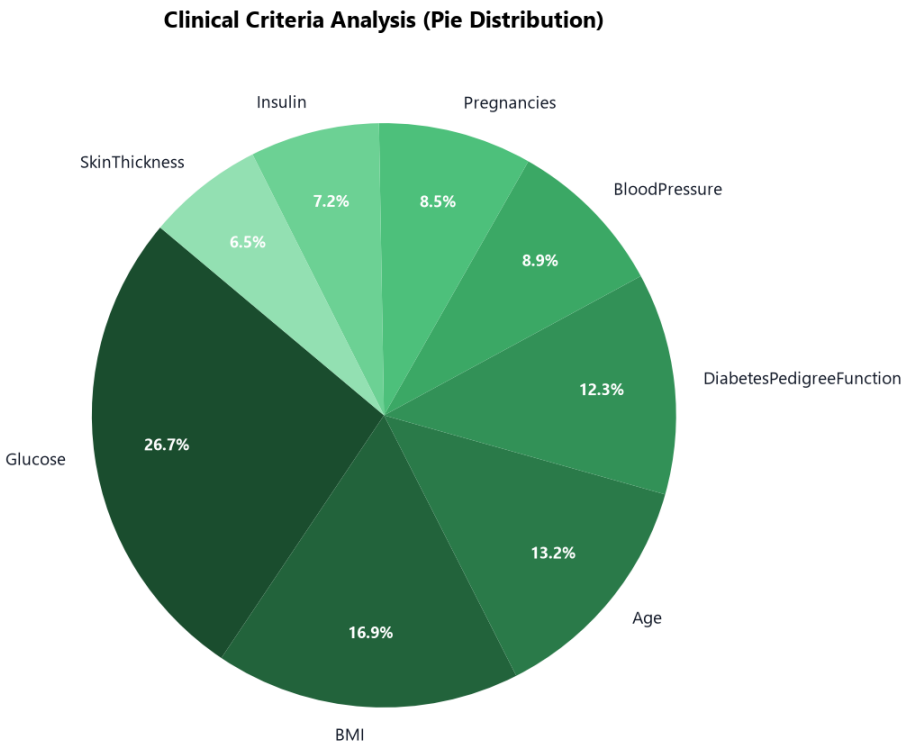


Figure (7-C): Circular Pie Chart - Depicting proportional relative weights

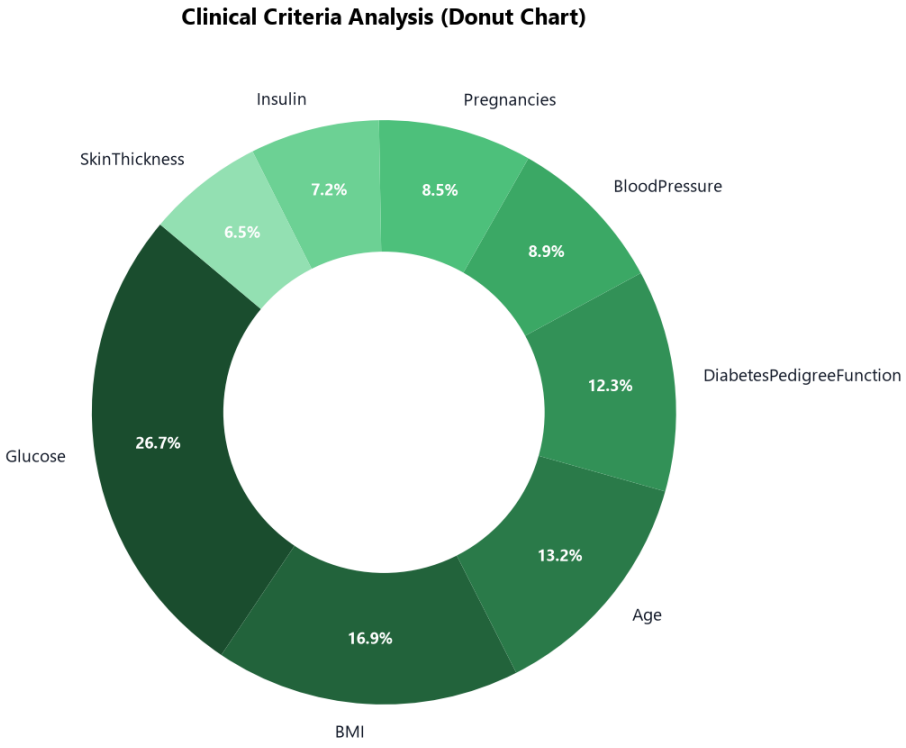


Figure (7-D): Donut Chart - Modern presentation of proportional weights

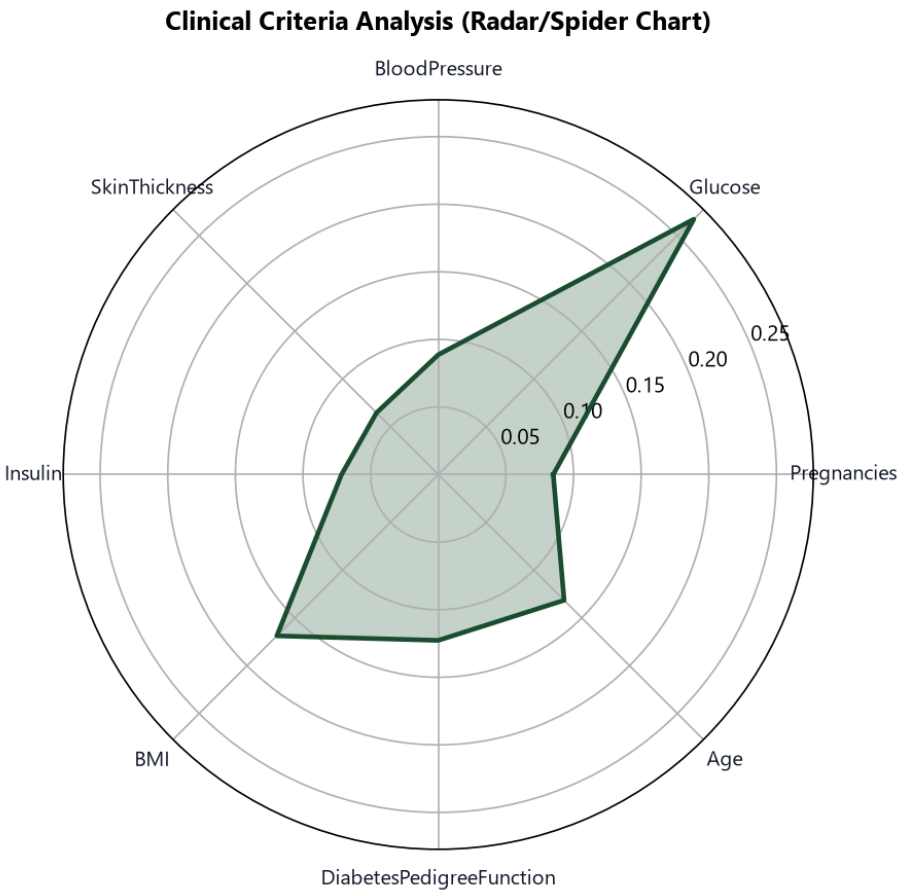


Figure (7-E): Radar / Spider Chart - Radial biomarker importance profile

Relative Mathematical Weights and Contribution Breakdown (Available in collapsible expander table):

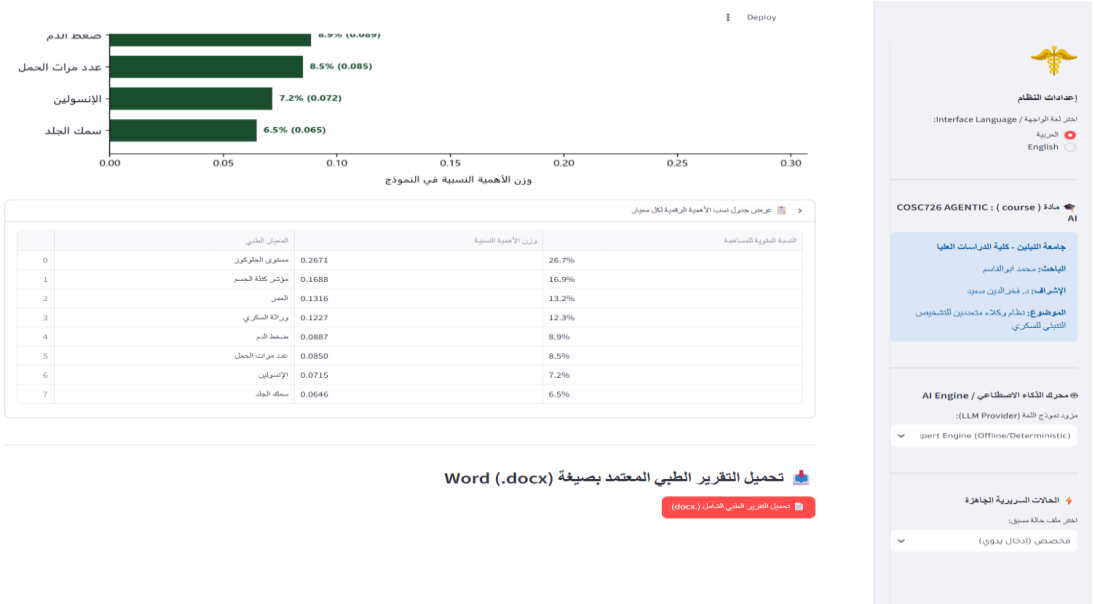


Figure (8): Collapsible Numerical Data Table displaying exact Relative Weights and Contribution Percentages

Clinical Criterion	Feature Identifier	Model Importance Weight	Contribution Percentage
Glucose Level	Glucose	0.2671	26.7%
Body Mass Index	BMI	0.1688	16.9%
Age	Age	0.1316	13.2%
Diabetes Pedigree Function	DiabetesPedigreeFunction	0.1227	12.3%
Blood Pressure	BloodPressure	0.0887	8.9%
Number of Pregnancies	Pregnancies	0.0850	8.5%
Serum Insulin	Insulin	0.0715	7.2%
Triceps Skin Thickness	SkinThickness	0.0646	6.5%

8. Tab 3: Lifestyle, Nutrition & Physical Activity Prescription

Tab 3 translates predictive analytics into actionable preventative and therapeutic lifestyle interventions. The Lifestyle & Nutrition Agent personalizes dietary guidelines, exercise regimens, and diagnostic monitoring based on patient status:

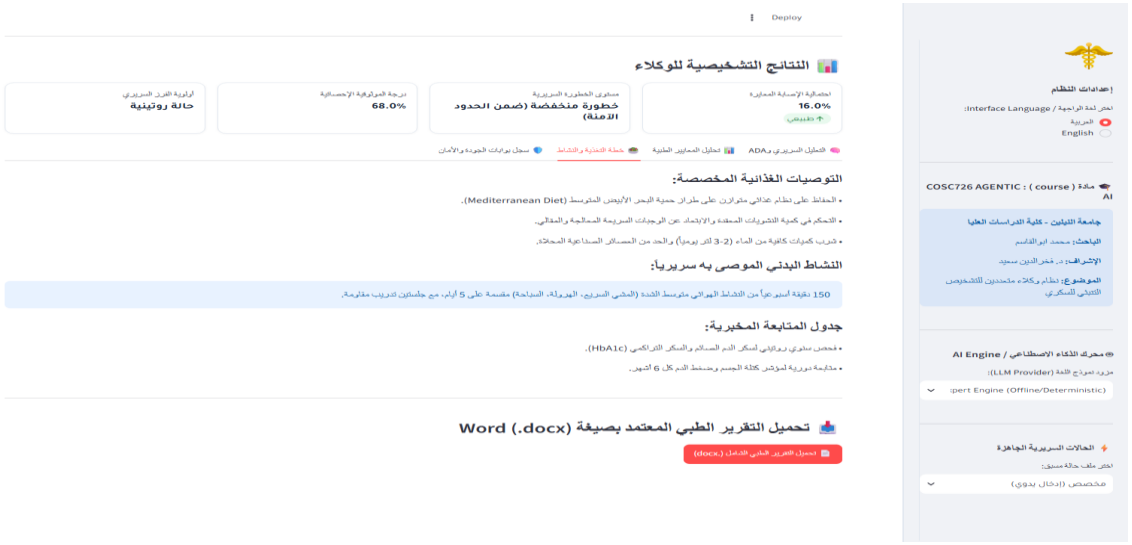


Figure (9): Tab 3 - Personalized Dietary Guidelines, Exercise Prescription, and Lab Follow-up Schedule

Key Pillars of Lifestyle Prescription:

1. Dietary Prescriptions: Emphasizing Mediterranean-style nutritional patterns, controlling refined carbohydrates, eliminating trans-fats, and maintaining 2-3 liters of daily hydration.
2. Clinical Physical Activity Prescription (Rendered in prominent blue container): 150 minutes per week of moderate-intensity aerobic exercise (brisk walking, jogging, swimming) across 5 days, supplemented with 2 resistance training sessions.
3. Diagnostic Follow-up Schedule: Annual fasting blood glucose and HbA1c testing, with bi-annual BMI and blood pressure assessments.

9. Tab 4: Safety Audit Trail & Quality Gates Verification

Tab 4 demonstrates regulatory compliance and governance. The Safety Gatekeeper Agent logs the validation outcome of all five quality gates, concluding with an official clinical and legal disclaimer:

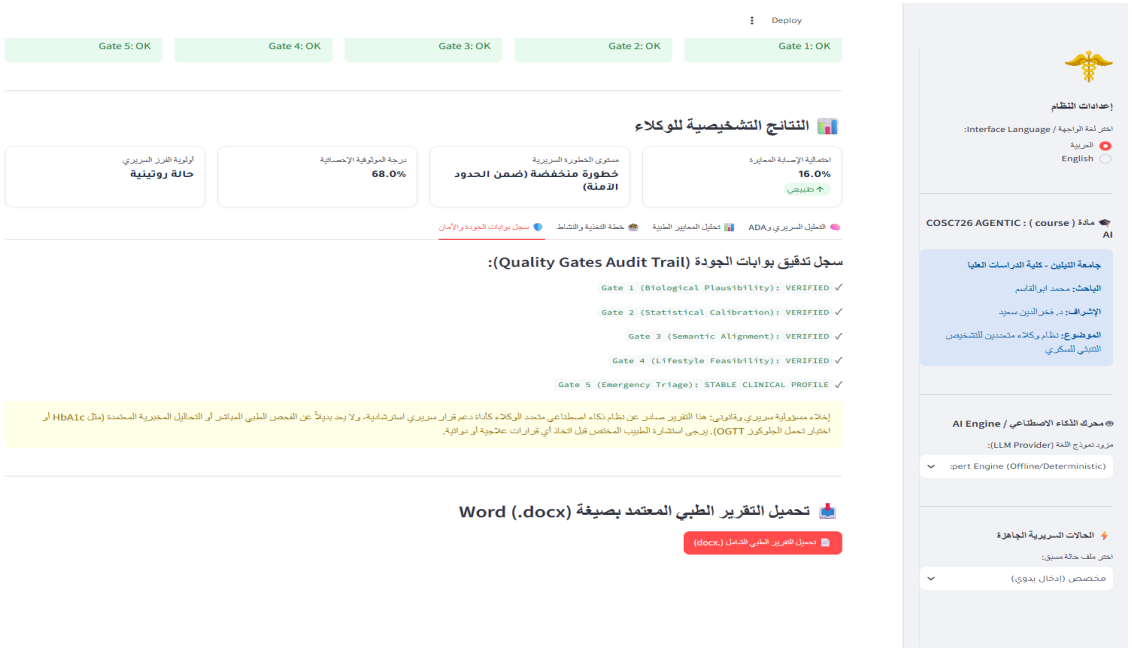


Figure (10): Tab 4 - Quality Gates Audit Trail and Official Clinical Safety Disclaimer

The 5 Audited Quality Gates:

• Gate 1 (Biological Plausibility): VERIFIED - Validated baseline inputs against physiological thresholds.
• Gate 2 (Statistical Calibration): VERIFIED - Confirmed calibration accuracy and confidence bounds.
• Gate 3 (Semantic Alignment): VERIFIED - Ensured semantic congruence between metrics and narrative.
• Gate 4 (Lifestyle Feasibility): VERIFIED - Verified safety and feasibility of lifestyle prescriptions.
• Gate 5 (Emergency Triage): STABLE CLINICAL PROFILE - Verified absence of acute decompensation.

[Official Clinical & Legal Safety Disclaimer]

Clinical & Legal Disclaimer: This report is generated by a multi-agent artificial intelligence decision support system for guidance purposes only. It does not replace direct physician examination or certified laboratory diagnostics (e.g. HbA1c, OGTT). Consult a qualified healthcare provider prior to making clinical or pharmacological decisions.

10. Critical Emergency Triage Interception System

The system incorporates an automated emergency tripwire mechanism triggered whenever acute, life-threatening hyperglycemia is detected (Glucose >= 250 mg/dL). The Safety Gatekeeper intercepts normal workflow, activates a high-priority red alert (#b91c1c), flags risks of Diabetic Ketoacidosis (DKA) or Hyperosmolar Hyperglycemic State (HHS), and instructs immediate emergency medical services contact:

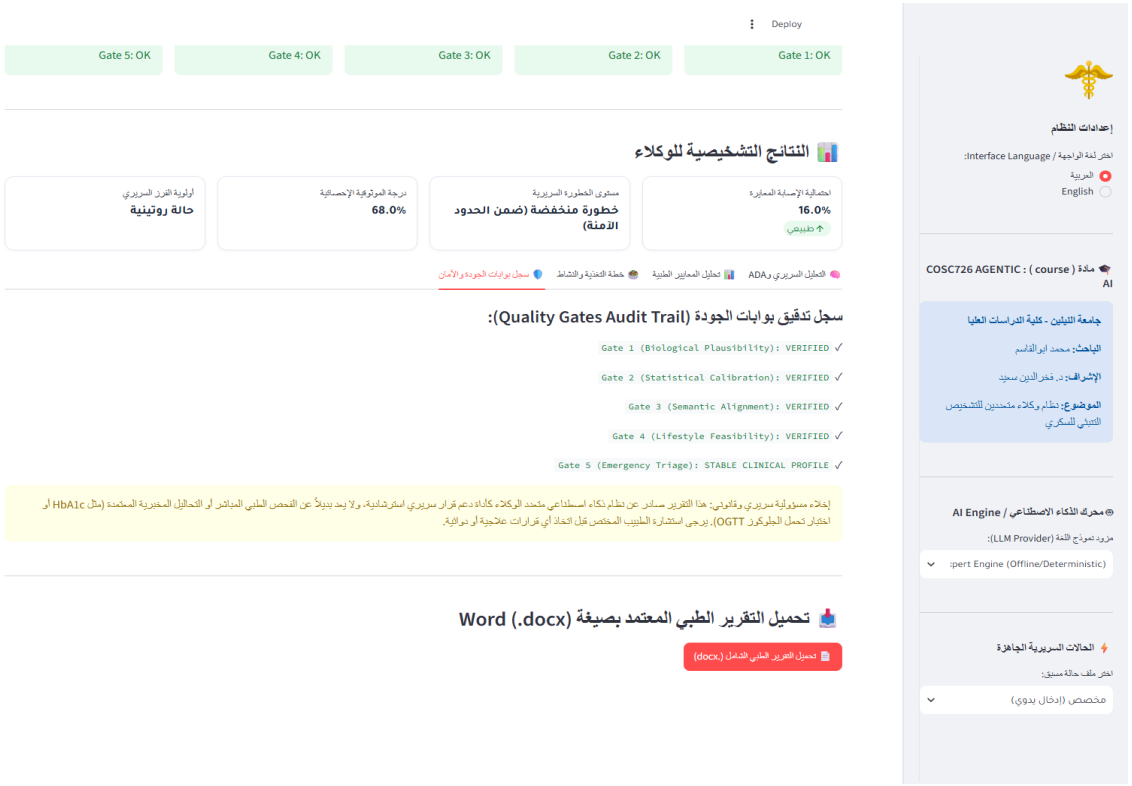


Figure (11): Real-time Critical Emergency Alert Banner triggered upon Glucose >= 250 mg/dL

[Acute Emergency Clinical Protocol Active]

CRITICAL CLINICAL EMERGENCY: Plasma Glucose has reached 260 mg/dL (exceeding emergency threshold of 250 mg/dL). Emergency clinical protocol is active. Seek immediate emergency medical care or call an ambulance to prevent acute metabolic decompensation.

11. Official Patient Clinical Report Download (.docx)

The dashboard enables one-click generation of comprehensive, print-ready Microsoft Word (.docx) clinical reports. To uphold medical privacy and clinical standards, patient reports are strictly sanitized of all academic metadata (author, supervisor, and course data are excluded), ensuring a purely professional medical document:

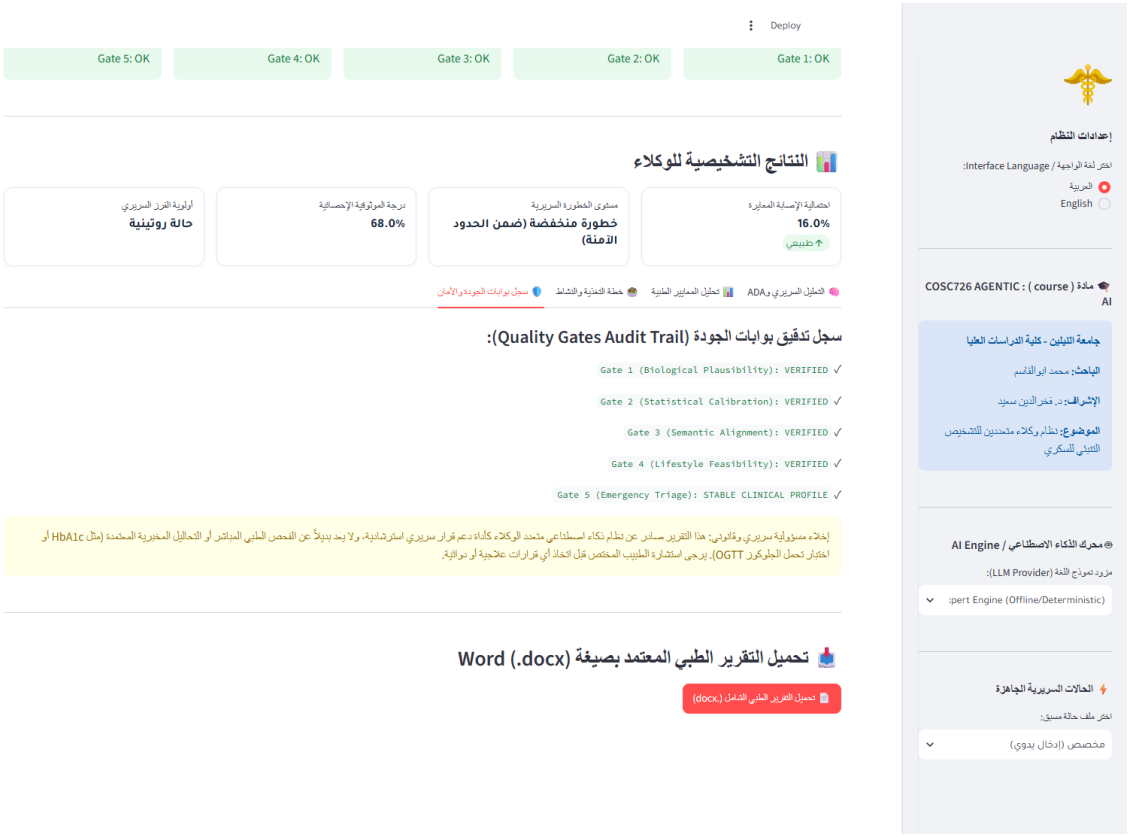


Figure (12): Official Patient Word Report (.docx) One-Click Download Section

Specifications of Generated Clinical Reports:

- Encrypted Patient Identifier (e.g. PAT-2026-09) protecting privacy.
- Tabular physiological metrics compared against clinical reference standards.
- Statistical risk evaluation and verified clinical diagnosis.
- Complete lifestyle, nutrition, and exercise prescription.
- Physician signature line, certification seal, and mandatory safety disclaimer.

12. Quick Start & CLI Master Controller

The upgraded project can be launched and verified using the following execution modes:

1. One-Click Batch Launchers:

- Double-click `run_new_project.bat` in the root workspace.
- Or double-click `run_web_app.bat` inside `New_Project_After_Agents/`.

2. Direct Command Line (Streamlit):
```bash
cd New_Project_After_Agents
streamlit run app.py
```

3. Master Controller Command Suite:
```bash
python run_system.py --ui                  # Launches interactive web dashboard
python run_system.py --phase all           # Executes full test suite across all 4 phases (100% Passed)
python run_system.py --presentation        # Re-generates bilingual master PowerPoint slide decks
python run_system.py --generate-all-reports   # Re-generates all 10 official Word documentation reports
```